

Introduction to Loops

Understanding Iteration using Flowcharts

Programming Fundamentals Team

SETU

2026-09-30

Outline

1. Why do we need loops?
2. Selection versus Iteration
3. Anatomy of a loop
4. Infinite loops
5. Useful loop patterns
6. Activities

Outline

1. Why do we need loops?

2. Selection versus Iteration

3. Anatomy of a loop

4. Infinite loops

5. Useful loop patterns

6. Activities

1.1 What if we needed to repeat this?

Without loops, repeated tasks quickly become messy:

```
print("Welcome")  
print("Welcome")  
print("Welcome")  
print("Welcome")
```

A loop lets us write the repeated instruction once.

1.1 What if we needed to repeat this?

Key idea

Computers are very good at repetition.

1.2 Everyday repetition

- Checking each student's mark
- Taking orders until the queue is empty
- Asking for a password until it is correct
- Displaying a menu until the user chooses Exit
- Reading temperatures from a sensor

Loops are used when a set of instructions must be repeated.

1.3 What is iteration?

Iteration means repeating a set of instructions until a condition is met.

Examples:

- Continue asking until the input is valid.
- Continue counting until the target is reached.
- Continue processing until there is no more data.

Outline

1. Why do we need loops?

2. Selection versus Iteration

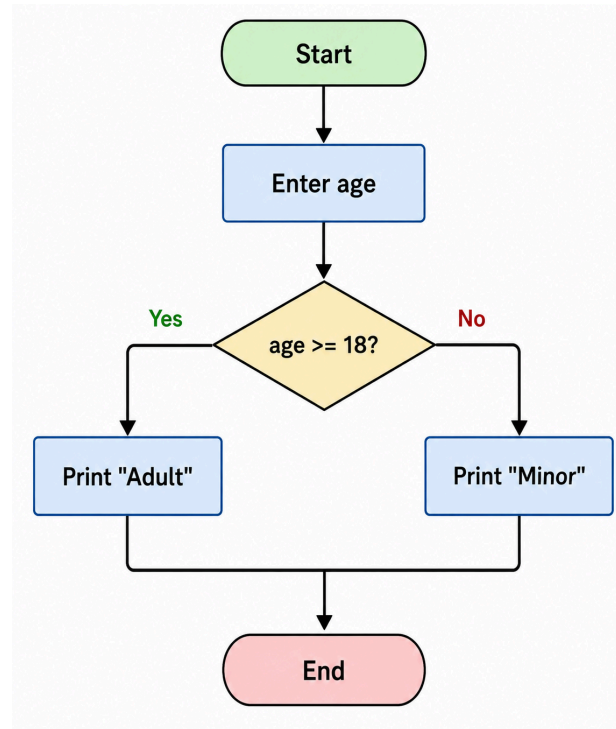
3. Anatomy of a loop

4. Infinite loops

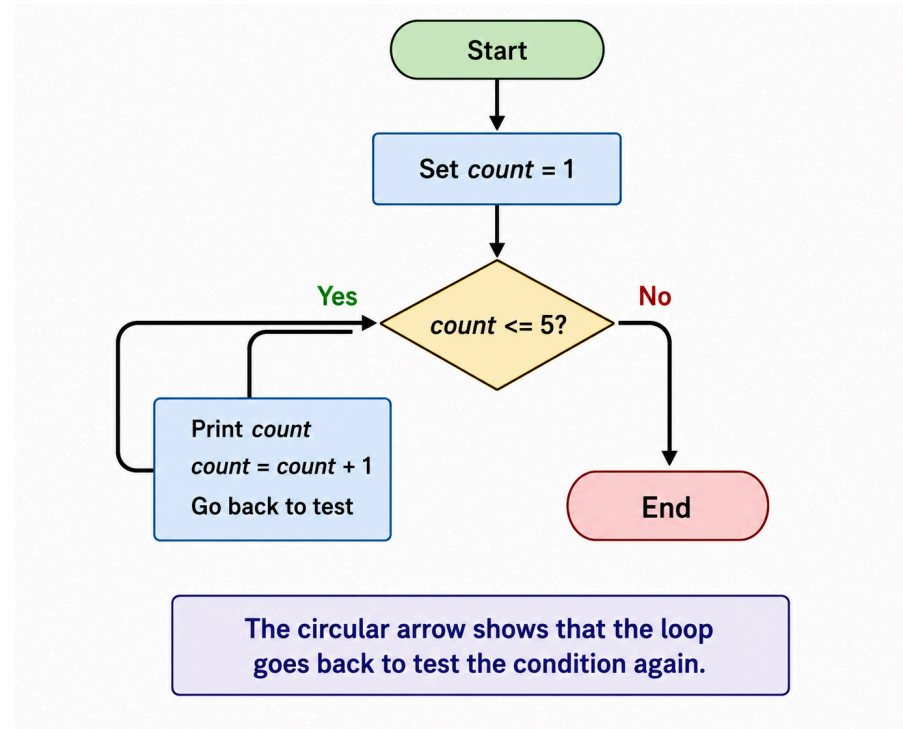
5. Useful loop patterns

6. Activities

2.1 Selection asks once



2.2 Iteration asks repeatedly



2.3 The key difference

Selection

- Condition checked once
- One path is chosen
- Program continues

2.3 The key difference

Iteration

- Condition checked repeatedly
- Instructions may run many times
- Stops when condition becomes false

Outline

1. Why do we need loops?
2. Selection versus Iteration
- 3. Anatomy of a loop**
4. Infinite loops
5. Useful loop patterns
6. Activities

3.1 Every loop needs four things

1. Starting value



Example: `count = 1`

2. Condition



Example: `count <= 5`

3. Repeated action



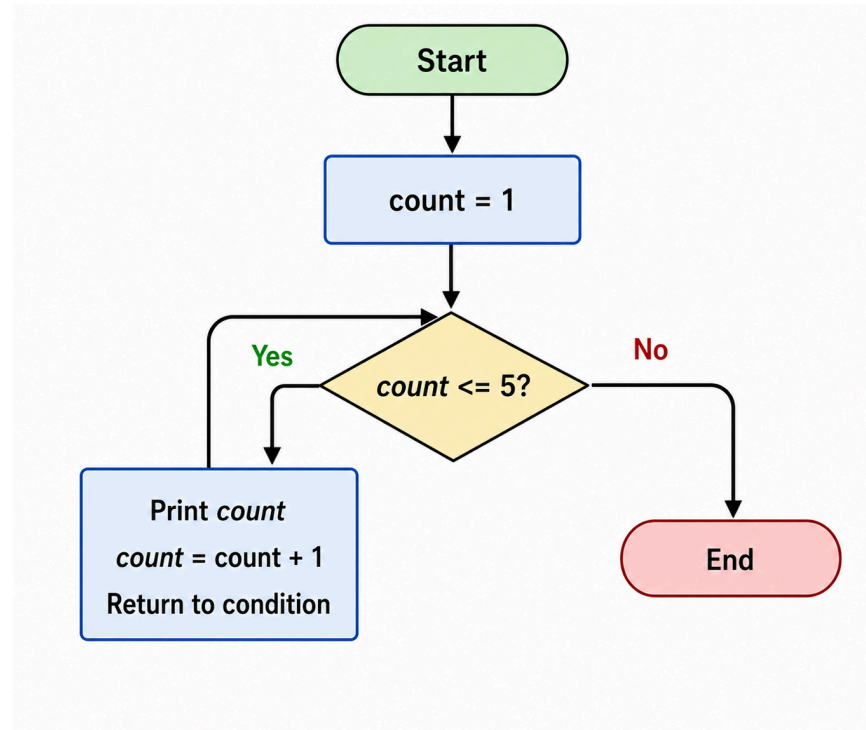
Example:
print the number

4. Update / change



Example:
`count = count + 1`

3.2 Counting to five: flowchart



3.3 Trace the loop

count	Condition	Output
1	1 <= 5 is True	1
2	2 <= 5 is True	2
3	3 <= 5 is True	3
4	4 <= 5 is True	4
5	5 <= 5 is True	5
6	6 <= 5 is False	Stop

3.4 Why does the loop stop?

The loop stops because the value changes.

count eventually becomes 6.

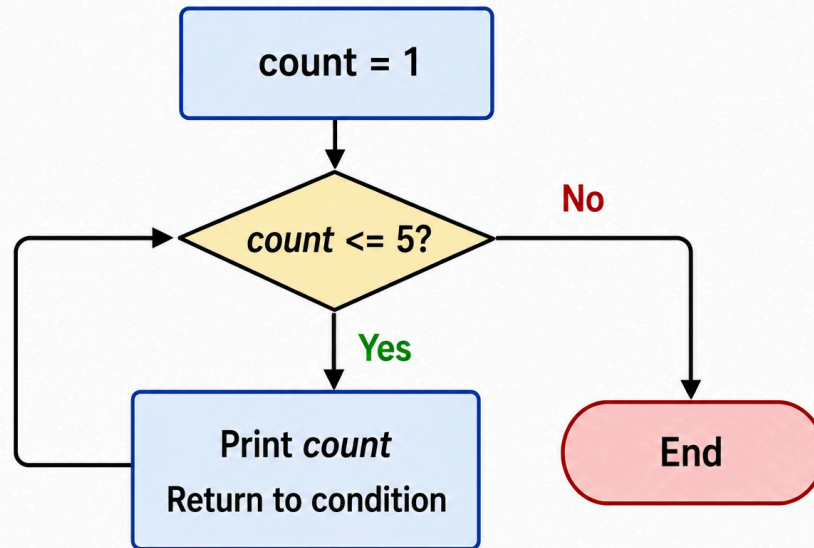
The condition `count <= 5` becomes false.

Important: a loop normally needs something inside it that moves it closer to stopping.

Outline

1. Why do we need loops?
2. Selection versus Iteration
3. Anatomy of a loop
- 4. Infinite loops**
5. Useful loop patterns
6. Activities

4.1 What if we forget the update?



count never changes, so the loop never ends.

4.2 Infinite loop in code

```
count = 1

while count <= 5:
    print(count)
```

Problem: `count += 1` is missing.

4.3 Corrected code

```
count = 1

while count <= 5:
    print(count)
    count += 1
```

Outline

1. Why do we need loops?

2. Selection versus Iteration

3. Anatomy of a loop

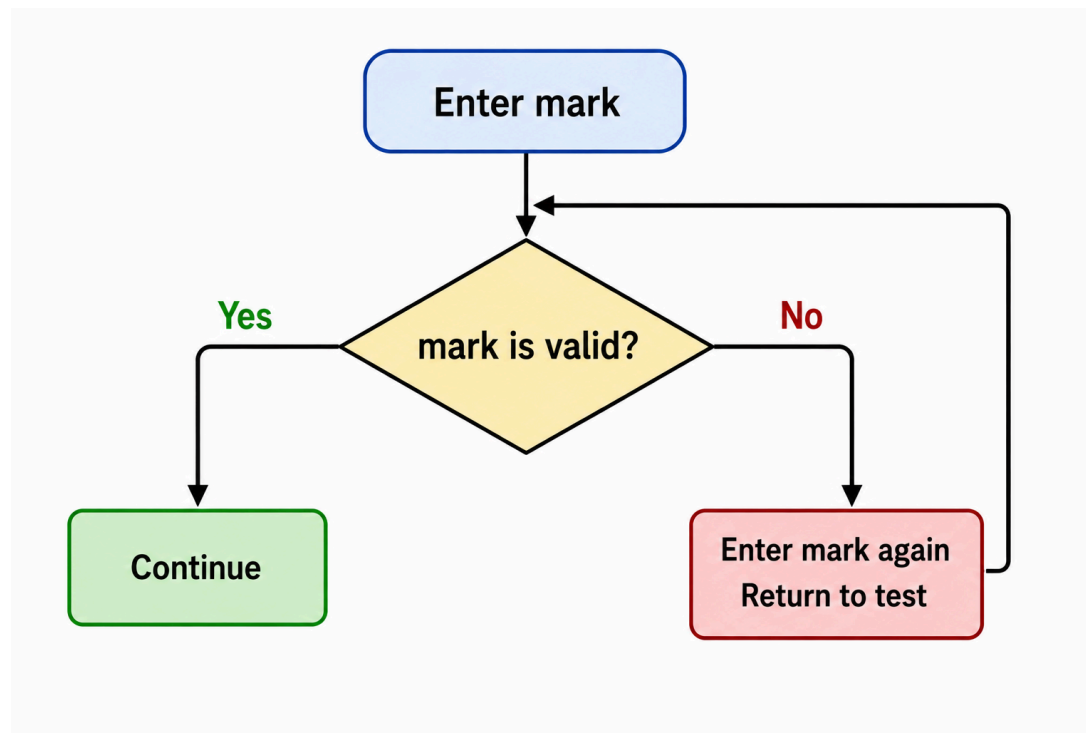
4. Infinite loops

5. Useful loop patterns

6. Activities

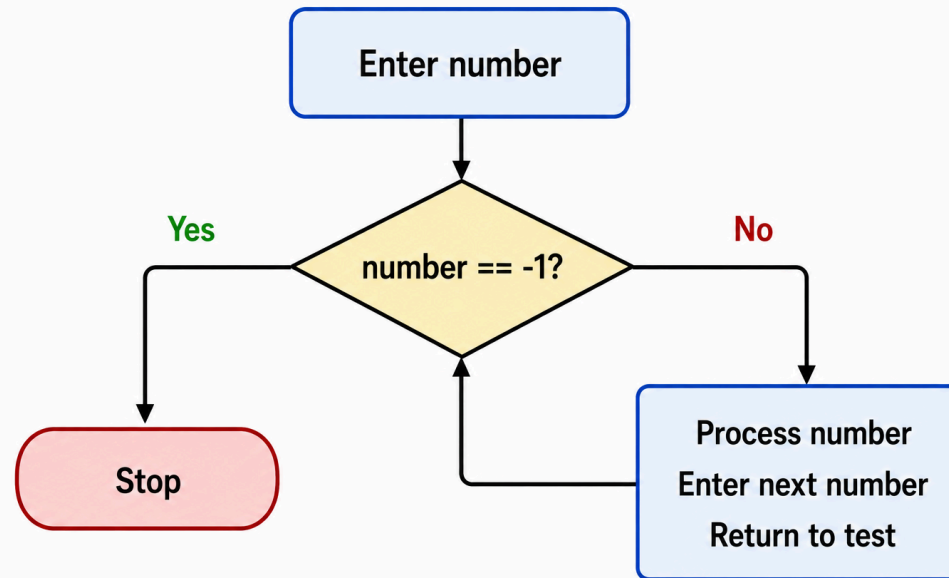
5.1 Validation loop

Use a loop when the user must enter valid data.



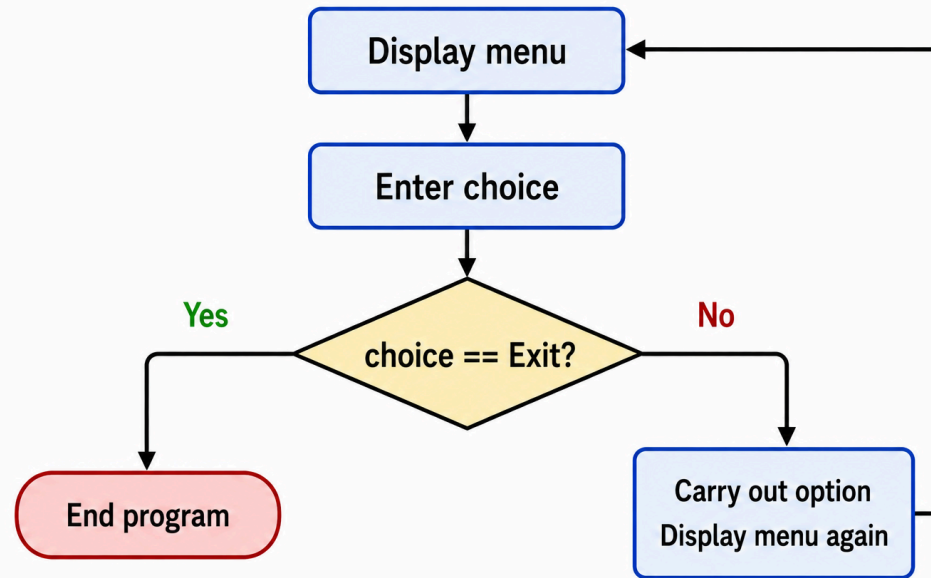
5.2 Sentinel loop

A **sentinel** is a special value that means “stop”.



5.3 Menu loop

Many programs repeat a menu until the user chooses **Exit**.



Outline

1. Why do we need loops?

2. Selection versus Iteration

3. Anatomy of a loop

4. Infinite loops

5. Useful loop patterns

6. Activities

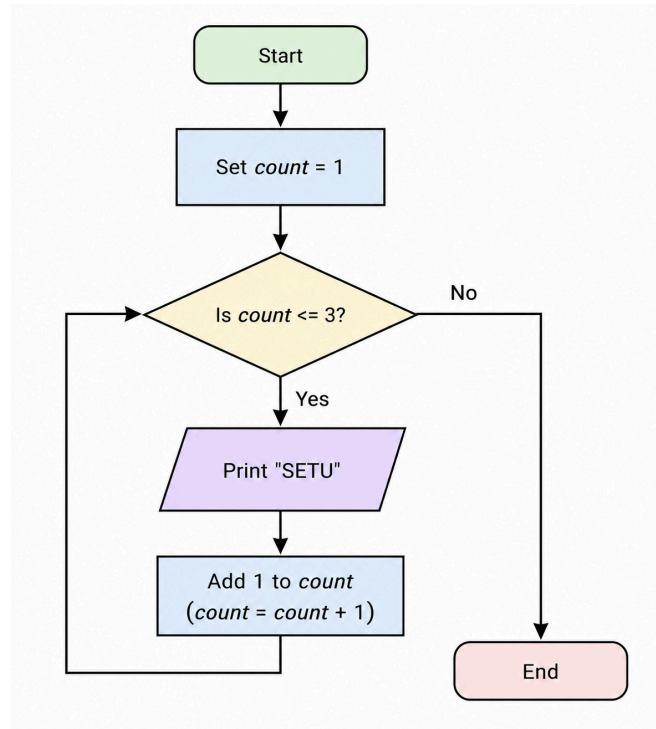
6.1 Activity: find the four parts

For the flowchart, identify:

1. Starting value
2. Condition
3. Repeated action
4. Update or stopping action

This prepares you to write the Python version in the next lecture.

6.2 Flowchart



6.3 Predict the output

Flowchart summary:

- Set count = 1
- While count <= 3
- Print "SETU"
- Add 1 to count

Question: what is printed?

6.4 Summary

- Iteration means repetition.
- Flowcharts help us plan loops before writing code.
- Selection asks once; iteration asks repeatedly.
- Every loop needs a condition and a way to stop.
- Missing updates can cause infinite loops.

Next: implementing these ideas in Python using `while` loops.

Thanks for Watching - Any questions?

